

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME:** Database Management Systems  
**COURSE CODE:** CSA05

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### COURSE OUTCOME COVERED

**CO3:** *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)  
Assessment Tool Weightage: 30%

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| QUESTIONS |                                                                                                                                                                                                           |                  | Total Marks: 50 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| Q.N<br>o  | Question                                                                                                                                                                                                  | Bloom's Level    | Marks           |
| 1         | Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies                 | BL2 – Understand | 5               |
| 2         | A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies | BL2 – Understand | 5               |
| 3         | Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF                                                                                                                  | BL2 – Understand | 5               |
| 4         | BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF                                                                                                                                            | BL2 – Understand | 5               |
| 5         | Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other                                                              | BL2 – Understand | 5               |
| 6         | Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly                                         | BL2 – Understand | 5               |
| 7         | A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An                                                           | BL2 – Understand | 5               |

|    |                                                                                                                                                                                             |                  |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---|
|    | attribute that depends on a non-prime attribute (d) A foreign key reference                                                                                                                 |                  |   |
| 8  | If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$ , then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined | BL2 – Understand | 5 |
| 9  | 5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form                                                            | BL2 – Understand | 5 |
| 10 | Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist                                | BL2 – Understand | 5 |

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**Code of Conduct:**

*I R.Pranathi(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:** *R. Pranathi*

## RUBRICS FOR CALCULATION:

| MCQ                      |               |                                                                       |                                              |                                                    |                                                     |
|--------------------------|---------------|-----------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| Criteria                 | Weightage (%) | Level 4 (Exemplary)                                                   | Level 3 (Proficient)                         | Level 2 (Developing)                               | Level 1 (Beginning)                                 |
| Conceptual Understanding | 30%           | Demonstrates thorough understanding; answers all questions accurately | Shows good understanding with few errors     | Partial understanding; several incorrect responses | Lacks understanding ; majority of answers incorrect |
| Accuracy of Answers      | 25%           | All answers are correct                                               | Most answers are correct with minor mistakes | Some correct answers; frequent errors              | Mostly incorrect answers                            |
| Application of Knowledge | 20%           | Correctly applies concepts to problem-based questions                 | Applies concepts with minor errors           | Limited ability to apply concepts                  | Unable to apply concepts                            |
| Time Management          | 15%           | Completes quiz well within time efficiently                           | Completes on time with minor delays          | Requires extra time or rushes through questions    | Unable to complete within given time                |
| Question Interpretation  | 10%           | Interprets all questions correctly and responds appropriately         | Misinterprets a few questions                | Often misinterprets questions                      | Unable to understand questions                      |

**1. Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies**

**ANS:**

(c) All attributes must be atomic

**2. A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies.**

**ANS:**

(b) Has no partial dependencies on primary key

**3. Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF**

**ANS:**

(d) 4NF

**4. BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF**

**ANS:**

( c ) 3NF

**5. Functional dependency  $X \rightarrow Y$  means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other**

**ANS:**

(b) X determines Y

**6. Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly**

**ANS:**

(d) Retrieval anomaly.

**7. A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference**

**ANS:**

(b) Any set of attributes that uniquely identifies tuples

**8. If  $R(A,B,C)$  has FD:  $A \rightarrow B$  and  $B \rightarrow C$ , then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined**

**ANS:**

(c) Transitively dependent on A

**9. 5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form**

**ANS:**

(c) Project-Join Normal Form

**10. Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist**

**ANS:**

(a) No data is lost during decomposition